

AI-Powered CRM + HRMS

Monolithic Module Architecture

1. Architecture Overview (Monolith)

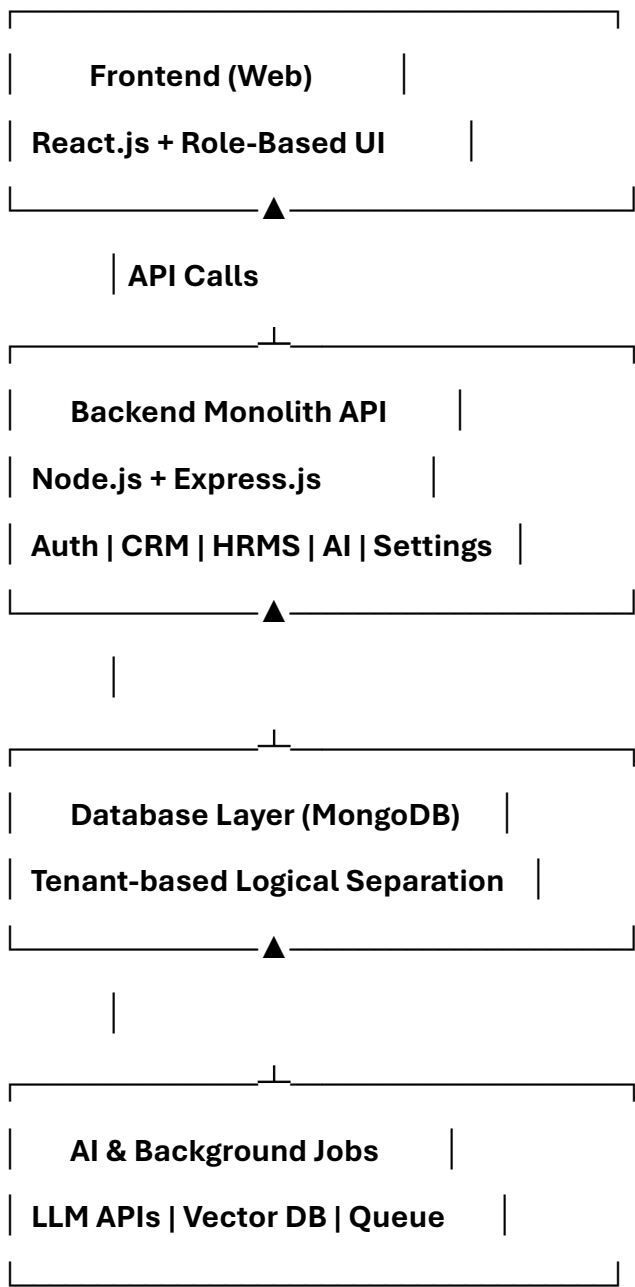
This system follows a **Modular Monolithic Architecture**, where:

- There is **ONE backend application**
- ONE database (logically separated by modules & tenants)
- All modules run inside the **same codebase**
- Modules are **functionally separated**, not physically separated (not microservices)

This approach ensures:

- Faster development
 - Easier deployment
 - Lower infrastructure cost
 - Strong data consistency
 - Easy AI integration across modules
-

2. High-Level System Layers



3. Core Monolithic Modules

3.1 Authentication & User Management Module

Purpose: Secure access & role control for the entire system.

Sub-Modules

- Login & Forgot Password (OTP based)
- User Management
- Role Mapping
- Session & Token Management (JWT)

Used By

- All modules (CRM, HRMS, AI, Settings)
-

3.2 Dashboard Module (Central Entry Point)

1. Admin Dashboard (System & Business Control)

Role

Company Owner / System Administrator

Responsibilities

- Overall business monitoring
- Security & compliance oversight
- High-level decision making
- AI usage & cost control

What Admin Can See

Key Widgets

- **Total Leads (All Sources)**
- **Total Customers**
- **Total Employees**
- **Active Users**
- **AI Alerts (Critical / High Priority)**
- **AI Credits Usage (Used vs Remaining)**

Charts & Analytics

- **Sales Funnel (Company-wide)**
- **Monthly Revenue Trend**
- **Employee Growth Trend**
- **Attendance Trend (Overall)**
- **AI Risk / Anomaly Chart**

Advanced Controls

- **Real-time security alerts**
- **SLA breach notifications**
- **System health status**
- **Downloadable reports (PDF / Excel)**
- **Custom KPI dashboard builder**

2. HR Dashboard (People Operations)

Role

HR Manager / HR Team

Responsibilities

- **Employee lifecycle management**
- **Attendance & leave monitoring**
- **Payroll accuracy**

- Talent performance tracking

What HR Can See

Key Widgets

- Total Employees
- Present vs Absent Today
- Pending Leave Requests
- Payroll Alerts (AI Detected)
- Performance Review Due

Charts & Analytics

- Attendance Trend (Daily / Monthly)
- Leave Utilization Trend
- Payroll Cost Trend
- Employee Performance Distribution
- Attrition Risk (AI Based)

AI Intelligence

- Attendance anomaly alerts
- Payroll fraud/error warnings
- AI leave approval recommendations
- Performance improvement suggestions

3. Sales Dashboard (Revenue & Pipeline Focus)

Role

Sales Executive / Sales Manager

Responsibilities

- Lead conversion
- Follow-ups & deals
- Revenue generation

What Sales Can See

Key Widgets

- **Total Leads (Assigned)**
- **Hot / Warm / Cold Leads (AI)**
- **Deals in Pipeline**
- **Today's Follow-ups**
- **Missed Follow-ups (AI Alert)**

Charts & Analytics

- **Sales Funnel (Personal / Team)**
- **Monthly Revenue Achievement**
- **Conversion Rate Trend**
- **Deal Win vs Loss Ratio**

AI Intelligence

- **Lead priority suggestions**
- **Deal close probability**
- **Best time to contact lead**
- **Auto follow-up reminders**

4. Manager Dashboard (Team Performance)

Role

Department Manager / Team Lead

Responsibilities

- **Team productivity**
- **Task & performance monitoring**
- **Approval & escalation handling**

What Manager Can See

Key Widgets

- **Team Members Count**
- **Team Attendance Status**

- **Team Leads / Tasks**
- **Pending Approvals (Leave / Sales)**
- **AI Risk Alerts (Team Level)**

Charts & Analytics

- **Team Performance Comparison**
- **Attendance Trend (Team)**
- **Sales / Productivity Output**
- **Goal vs Achievement**

AI Intelligence

- **Low productivity alerts**
 - **Performance risk indicators**
 - **Team workload imbalance detection**
-

Shared Widgets (Across All Dashboards)

- **Total Leads**
 - **Hot / Warm / Cold Leads (AI)**
 - **Employees**
 - **Pending Leaves**
 - **AI Alerts (Role-Filtered)**
-

Charts Engine (Dynamic & Role-Aware)

- **Sales Funnel**
 - **Monthly Revenue**
 - **Attendance Trend**
 - **Performance Trend**
 - **AI Risk / Anomaly Charts**
-

Advanced Dashboard Capabilities

- **Real-time updates (WebSocket based)**
- **Role-aware data filtering**
- **Custom KPI widgets**
- **Export reports (PDF / Excel)**
- **AI explainability (why alert was triggered)**

4. CRM Module (Sales Engine)

4.1 Lead Management Sub-Module

Handles the **entire lead lifecycle**.

Entities

- Lead
- Lead Activity
- Lead AI Score

AI Features

- Lead priority prediction
 - Hot/Warm/Cold classification
 - Consent compliance tracking
-

4.2 Sales Activities Sub-Module

Tracks all customer interactions.

Activities

- Calls
- Emails (tracked)
- WhatsApp
- Meetings

AI Enhancements

- Follow-up reminders
- Interaction analysis

4.3 Sales Pipeline & Forecasting

Pipeline

- Stage based (Kanban)
- Deal value
- AI close probability

Forecasting

- Monthly revenue prediction
- Expected vs Actual

4.4 Customer Management (Post-Sales) ★

Purpose: Convert leads into long-term customers.

AI Intelligence

- Renewal probability
- Relationship health score
- Churn risk

This module connects CRM → Support → Finance insights.

5. HRMS Module (People Operations Engine)

5.1 Employee Management

- Digital onboarding
- Profile management
- Salary structure

5.2 Attendance & Leave

Attendance

- Daily logs
- AI anomaly detection (fake punches, irregular hours)

Leave

- Requests & approvals
 - AI leave approval recommendation
-

5.3 Payroll Module

- Auto salary calculation
 - Payslip generation
 - AI payroll anomaly detection (fraud / mistakes)
-

5.4 Talent Intelligence (AI)

AI Resume Screening

- JD vs Resume match
- Skill gap analysis

AI Performance Review

- KPI analysis
 - Productivity score
 - Appraisal suggestions
-

6. Common AI Layer (Shared Across System)

This is **NOT a separate service**, but a **shared internal module**.

AI Capabilities

- AI Assistant (Chat)
- AI Smart Search (Natural Language)
- AI Notifications
- AI Fraud Detection
- AI Explainability Logs

Used By

- CRM

- HRMS
 - Dashboard
 - Admin Reports
-

7. Security & Compliance Module

Applies to **entire monolith**.

- RBAC (Role Based Access Control)
 - Field-level permissions
 - Data masking (Salary, PAN, GST)
 - Encryption key management
 - Audit logs
 - Backup & disaster recovery
-

8. System Settings Module

Global Configuration Layer

- Company profile
 - Roles & permissions
 - AI credits & usage tracking
 - Feature toggles (Plan based)
-

9. Database Design (Monolithic Schema)

Single database with **logical separation**:

Company

└─ Users

└─ Roles

└─ Leads

└─ Customers

└─ Employees

└— Attendance

└— Payroll

└— AI Logs

└— Audit Logs

Tenant isolation is handled using:

- companyId in every collection

10. Why Monolith is the Right Choice (Client Friendly)

- ✓ Faster MVP delivery
- ✓ Lower cloud cost
- ✓ Easier debugging
- ✓ AI works across all modules easily
- ✓ Can be converted to microservices later

11. Future Scalability (Optional)

- Extract AI module as service
- Extract Notification engine
- Introduce mobile apps
- Move heavy analytics to async workers

✓ FINAL RESULT

This is a **production-ready, enterprise-grade MONOLITH MODULE ARCHITECTURE** for an **AI-Powered CRM + HRMS**, fully aligned with your detailed module list and AI vision.